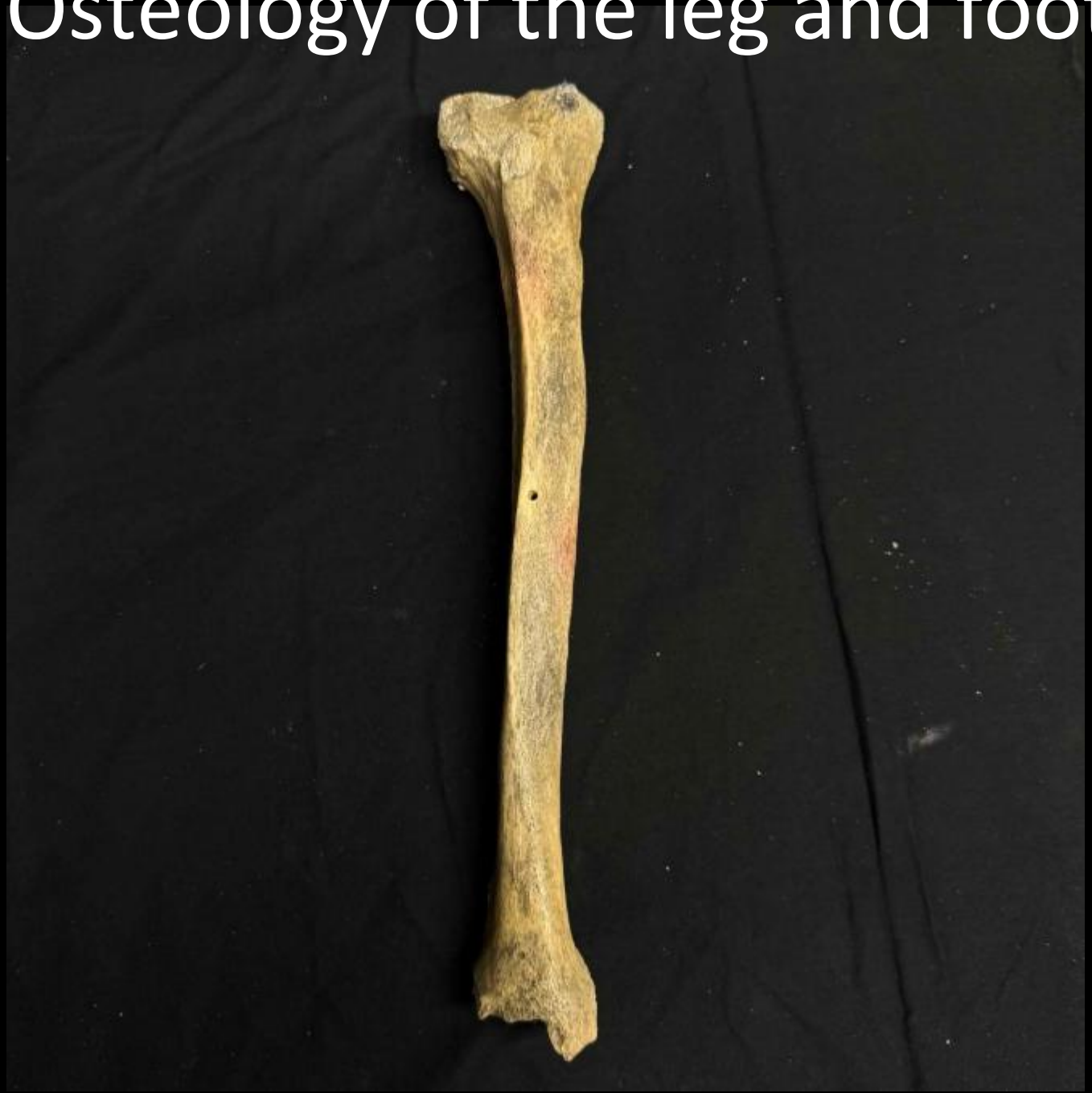
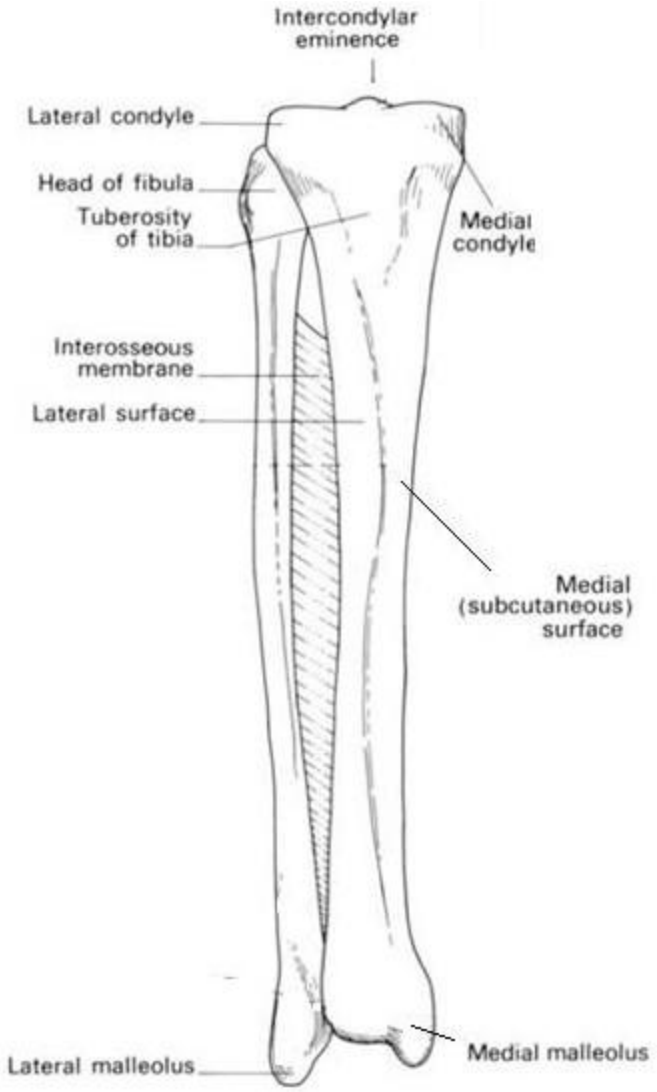
Osteology of the leg and foot



Learning Objectives

- Describe osteology of tibia and fibula
- Appreciate the significance of nutrient foramina of long bones
- State the law of ossification and know the one exception
- Revise knee and patellar anatomy
- Describe the inferior tibio-fibular syndesmosis
- Identify the tarsal bones
- List the medial and lateral ankle ligaments
- Consider the relevant clinical applications of anatomy
- List the four contributions of Percival Pott to surgery
- Classify ankle fracture severity using the Weber Scheme

Right Leg bones: anterior aspect

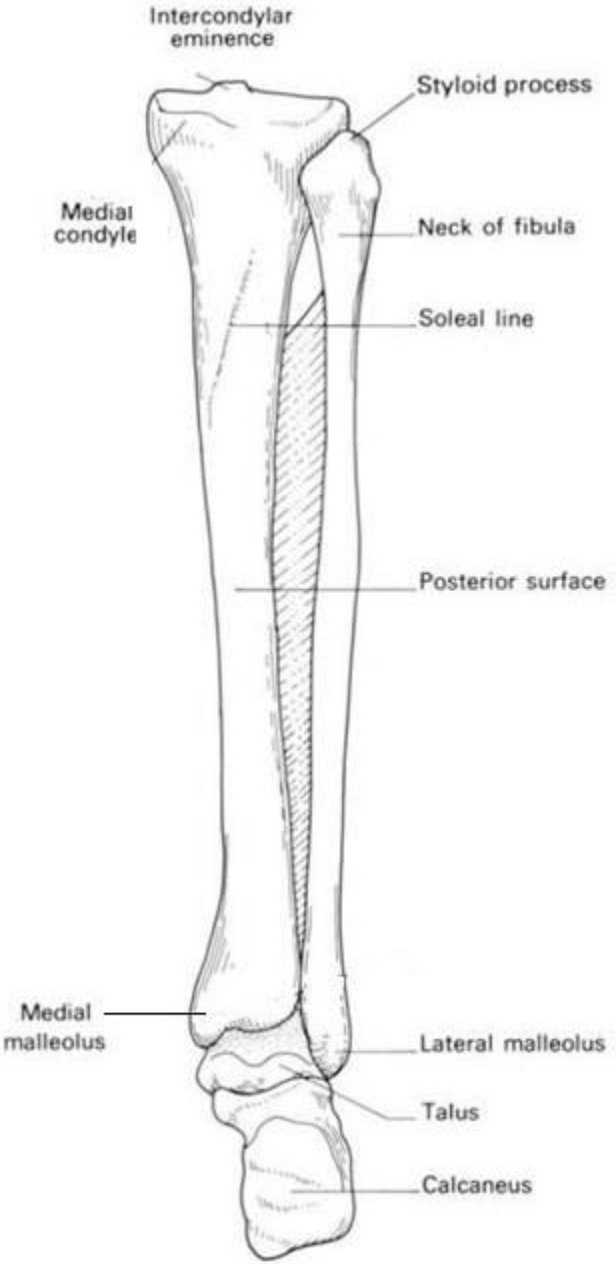


lateral

medial

Clinical Anatomy
Applied anatomy for students and junior doctors
Harold Ellis
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Right Leg bones: posterior aspect

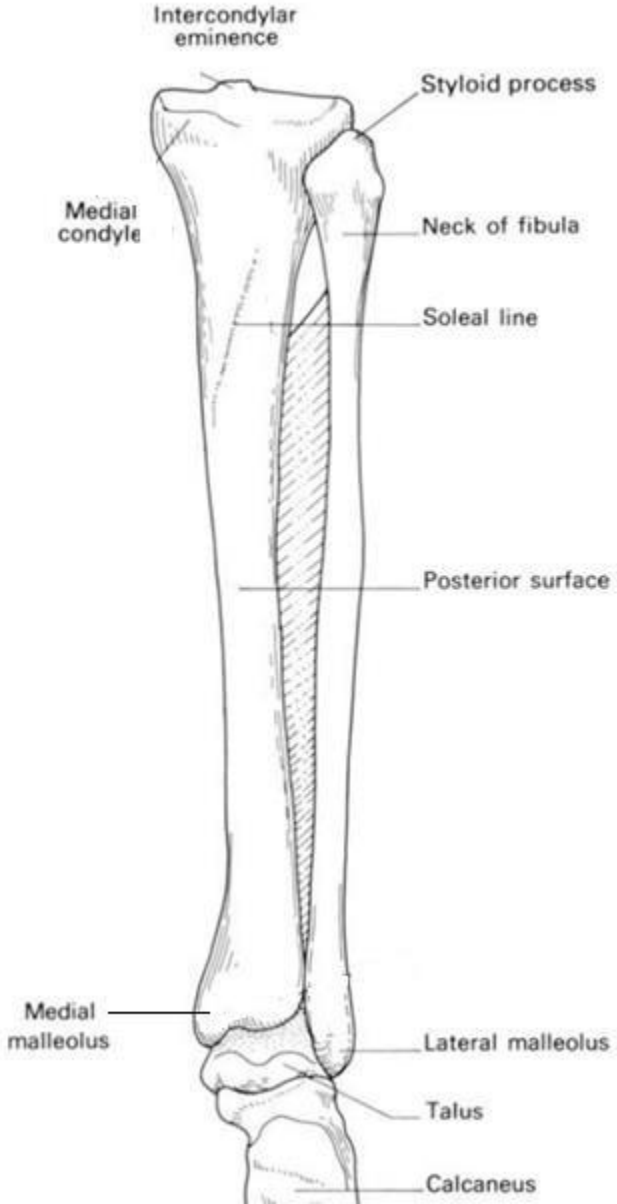


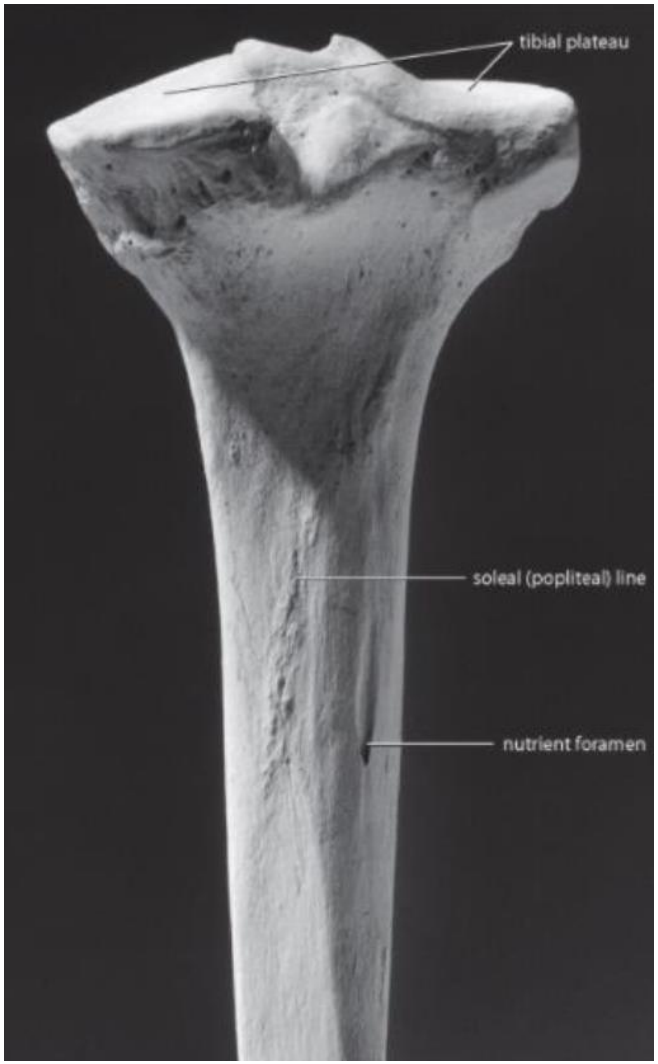
medial

lateral

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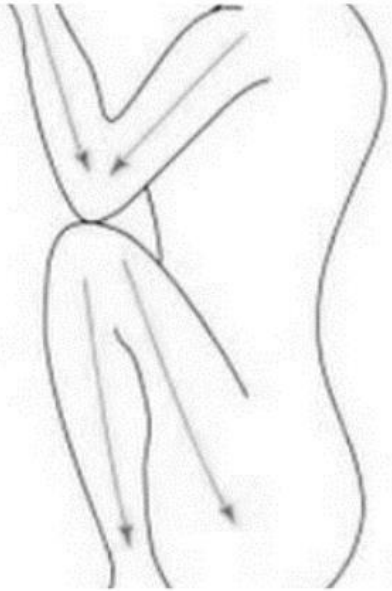
Nutrient artery of the tibia
Must be preserved during surgery



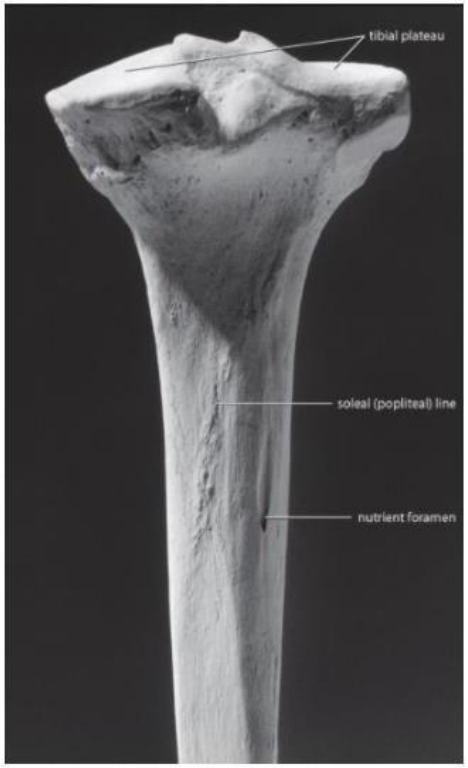


Direction of nutrient foramina in long bones

"From the knee I flee and to the elbow I grow"









radius

To elbow joint

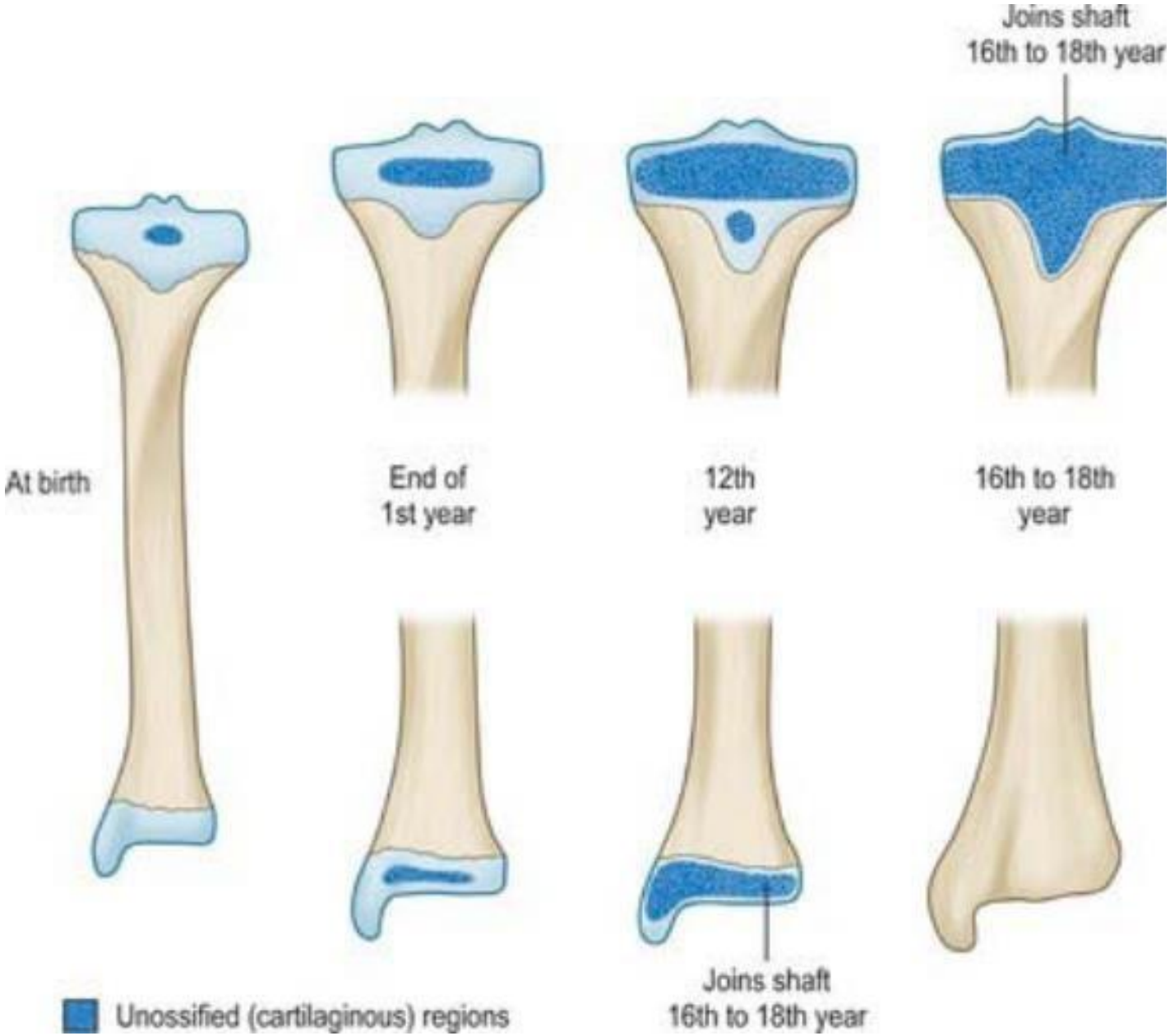


tibia

Away from knee joint

Study of Nutrient Foramina in Long Bones of Nigerians

U. Ukoha, Kosisochukwu E Umesiliugo, +3 authors, Izuchukwu F Obande · Published 2013 · Medicine, Biology · The Journal of medical research

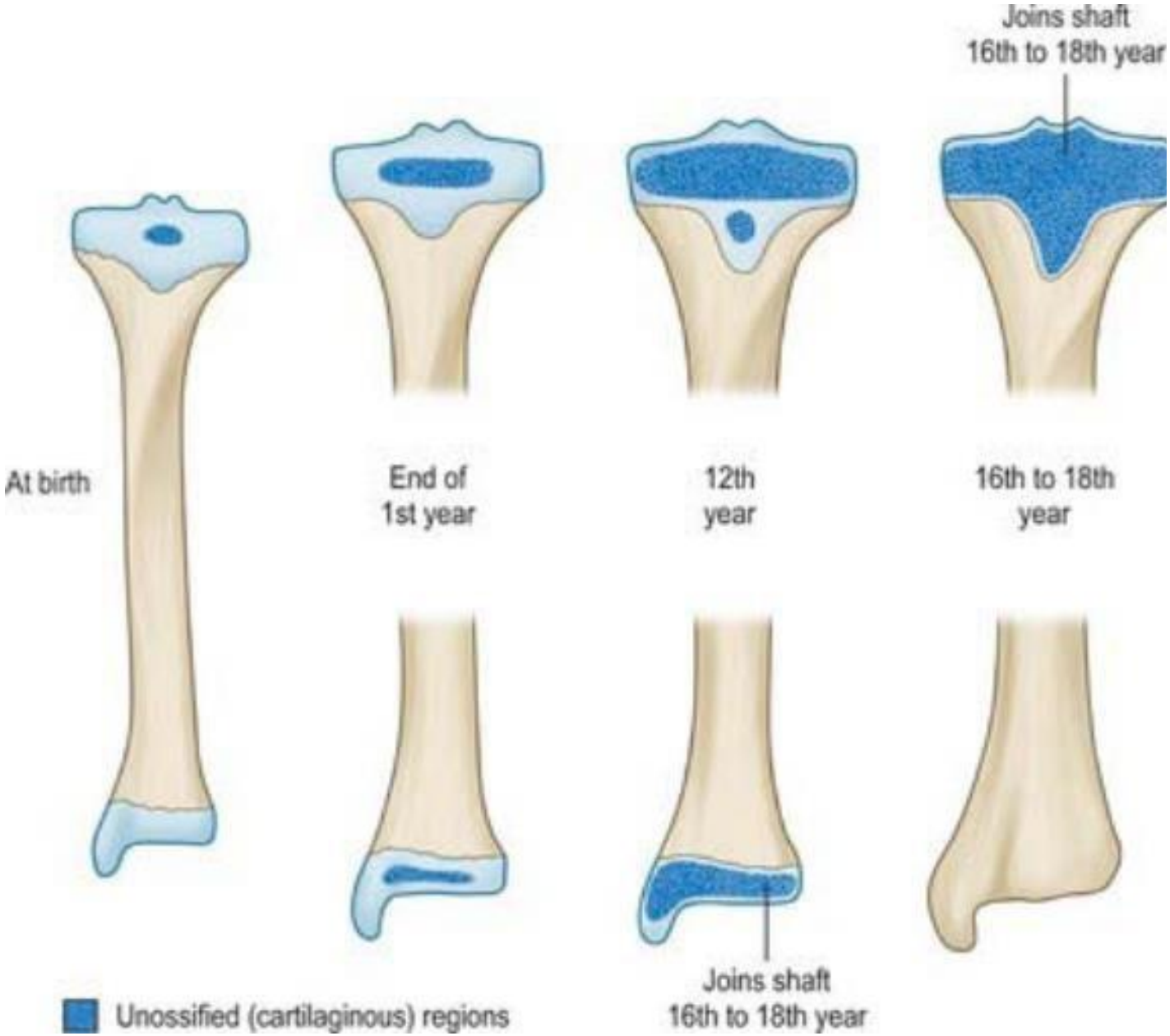

Joins shaft
16th to 18th year
At birth
End of
1st year
12th
year
16th to 18th
year
Joins shaft
16th to 18th year
Unossified (cartilaginous) regions

The Law of ossification:

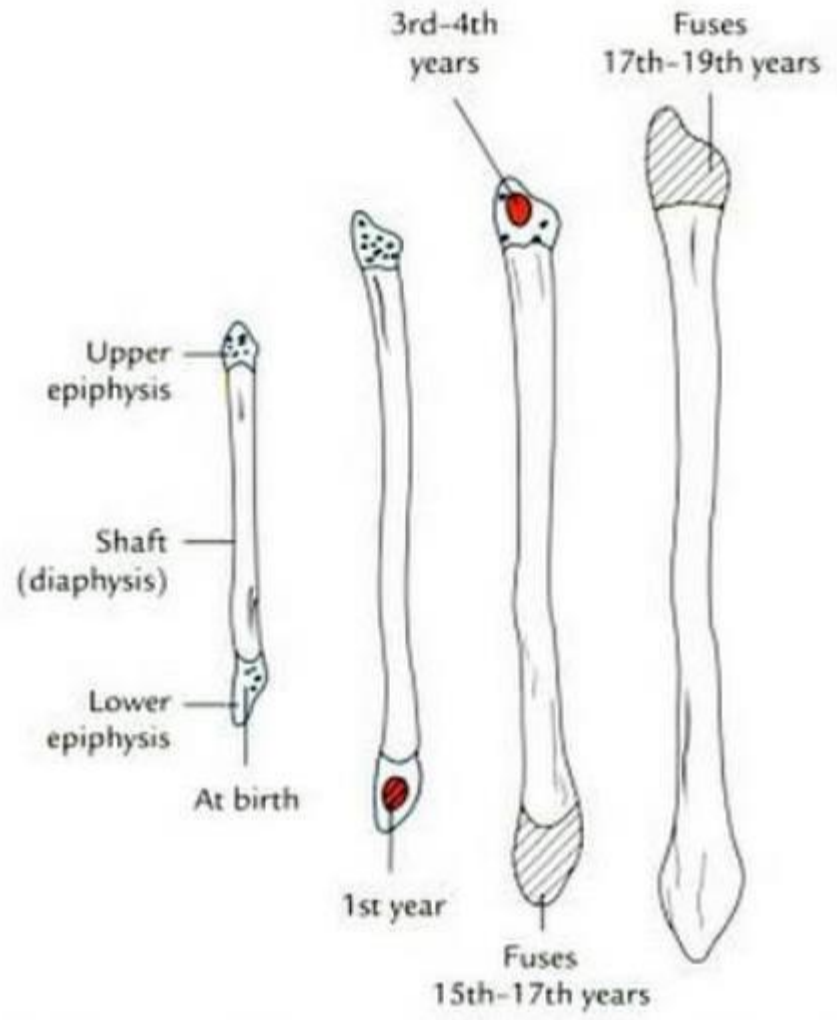
(Union of epiphyses)

The epiphyseal centre which

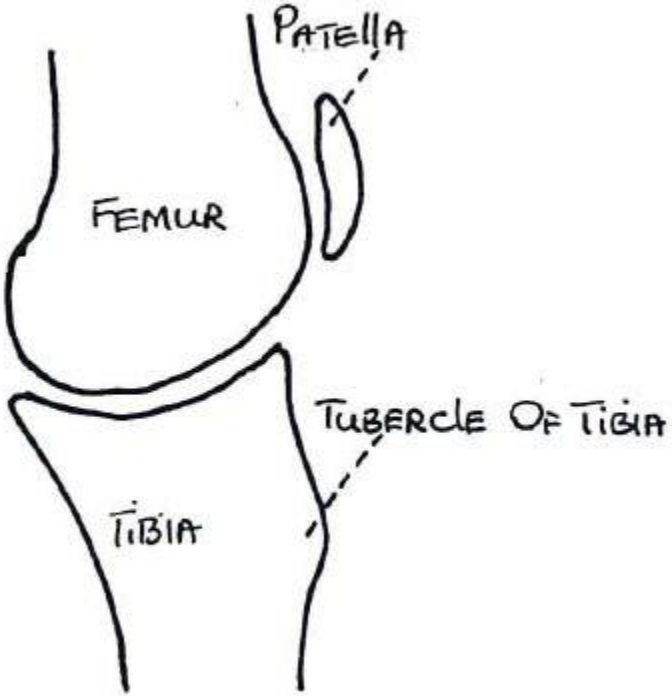
appears first……..**unites last**

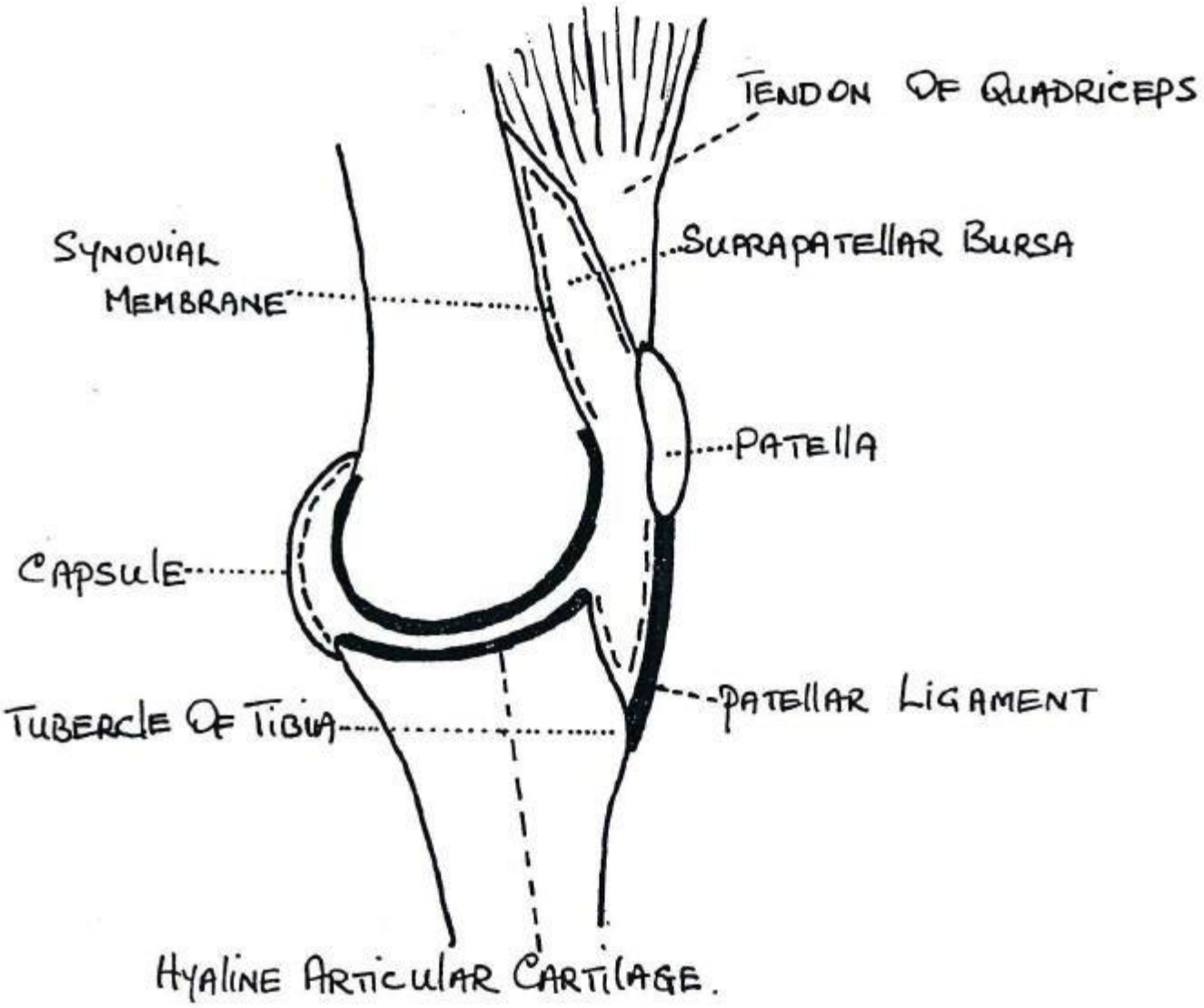

Joins shaft
16th to 18th year
At birth
End of
1st year
12th
year
16th to 18th
year
Joins shaft
16th to 18th year
Unossified (cartilaginous) regions

Only the head of the fibula violates the Law of ossification



Side view to show fit of bones of knee joint





Coakley Diagram

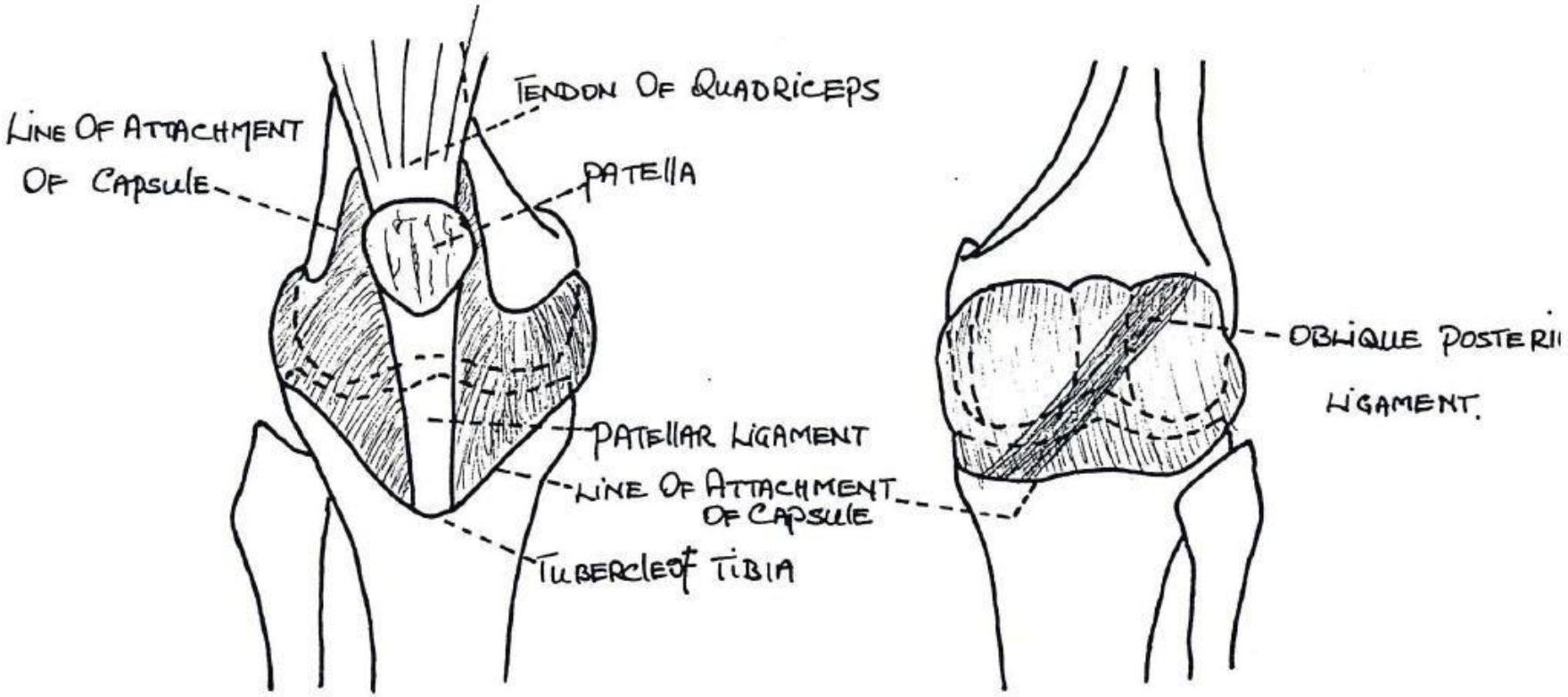
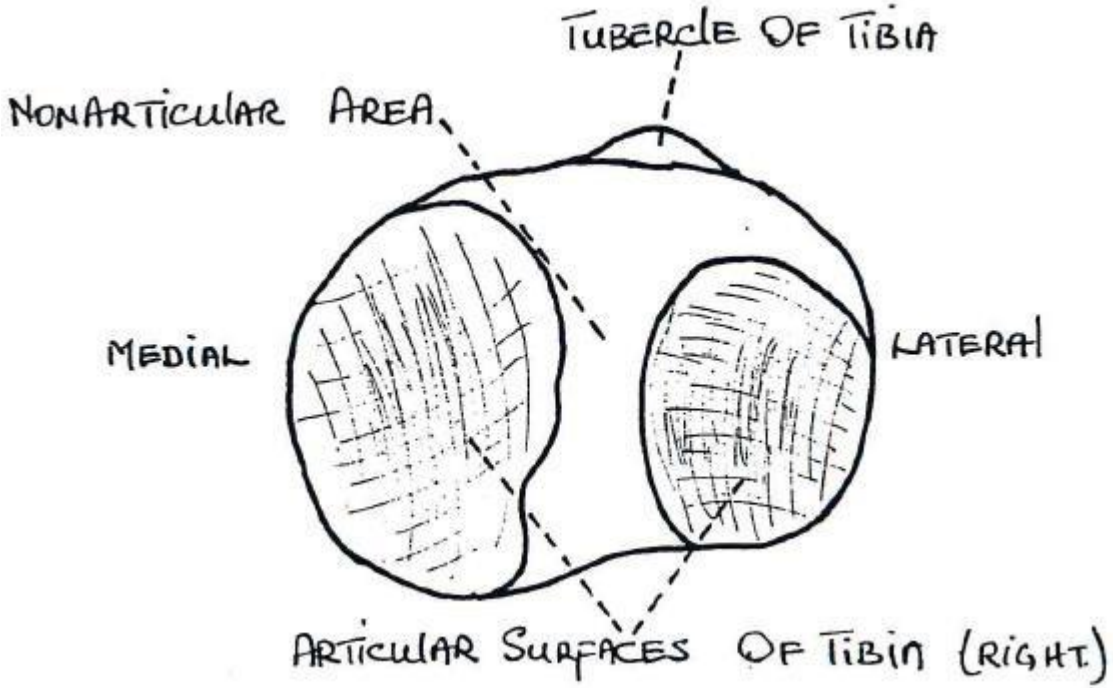
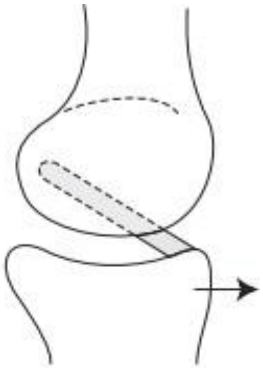

TENDON OF QUADRICEPS
LINE OF ATTACHMENT OF CAPSULE
PATELLA
PATELLAR LIGAMENT
LINE OF ATTACHMENT OF CAPSULE
TUBERCLE of TIBIA
OBLIQUE POSTERI
LIGAMENT.

Diagram to show the upper surface of right tibia

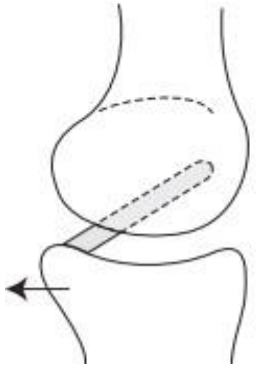


Side view of right Knee joint and cruciate ligaments

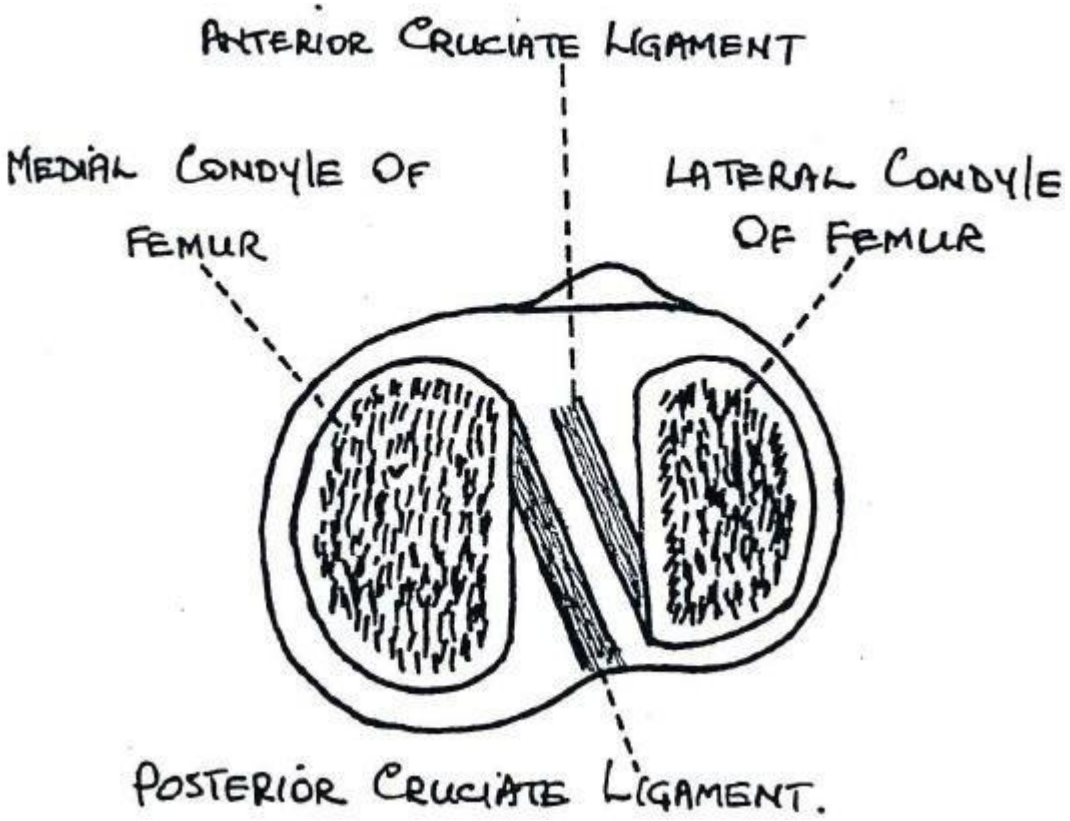


ACL prevents tibia slipping forward

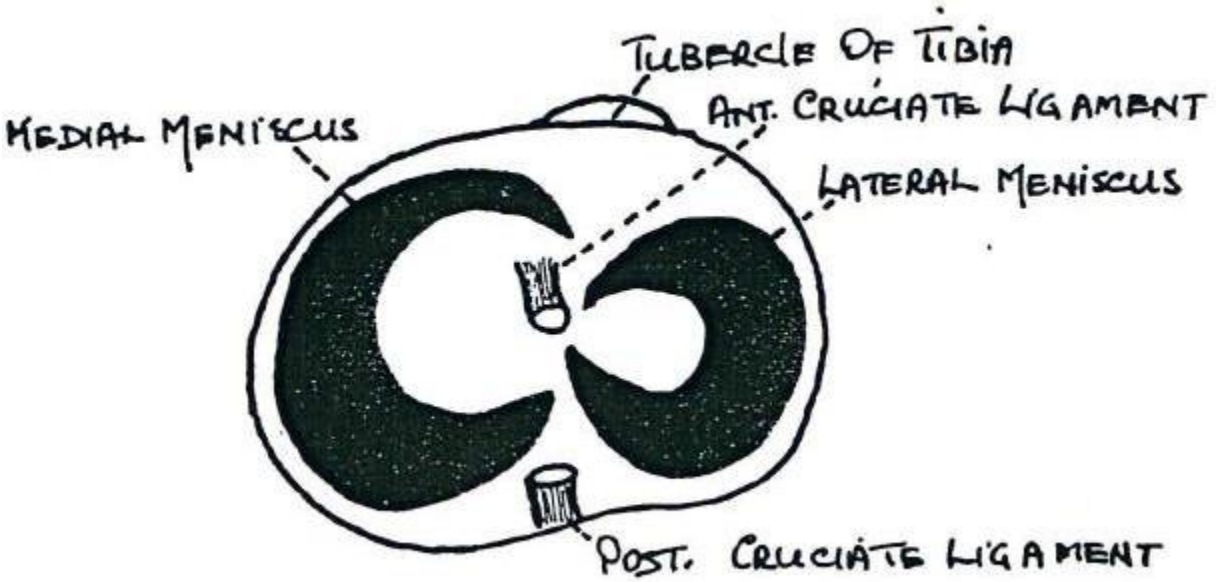
Posterior Anterior

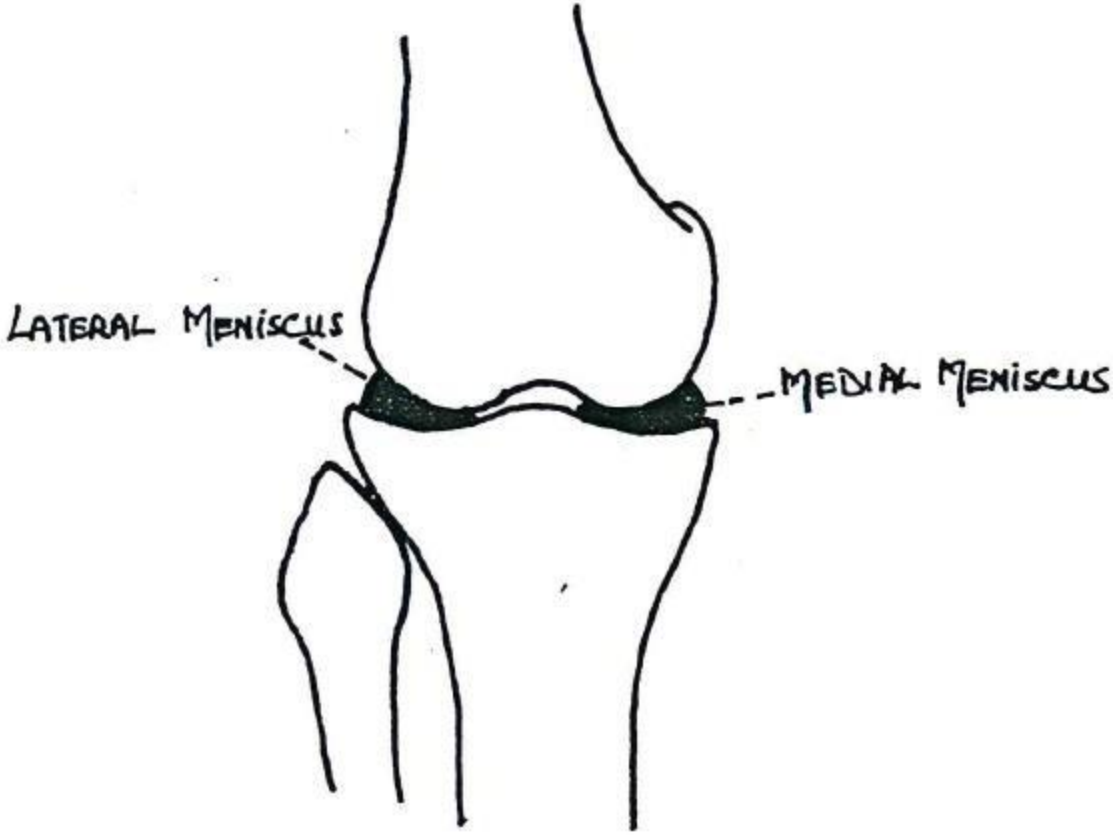


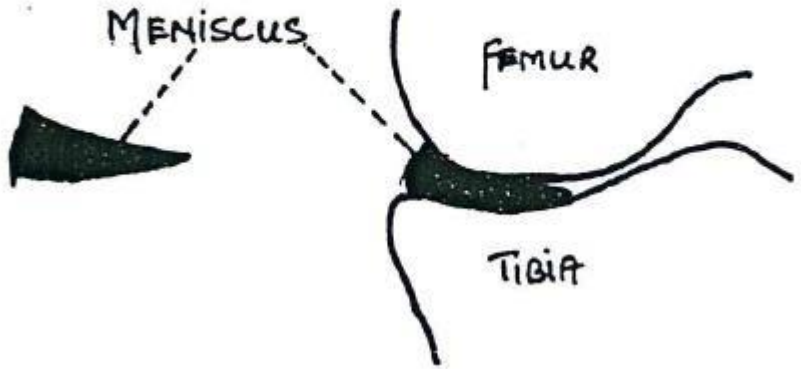
PCL prevents tibia slipping backwards

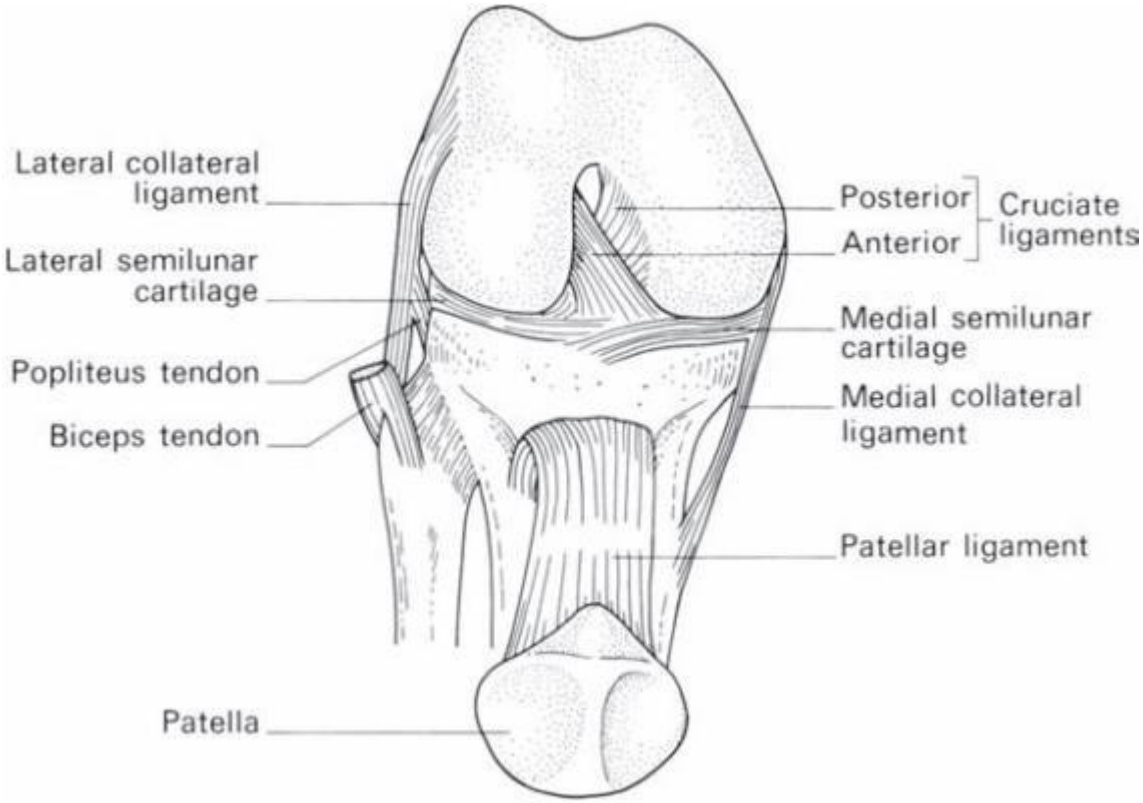


"LAMP" lateral anterior _ medial posterior


TUBERCLE OF TIBIA
ANT. CRUCIATE LIGAMENT
MEDIAL MENISCUS
LATERAL MENISCUS
POST. CRUCIATE LIGAMENT


LATERAL MENISCUS
MEDIAL MENISCUS


MENISCUS
FEMUR
TIBIA



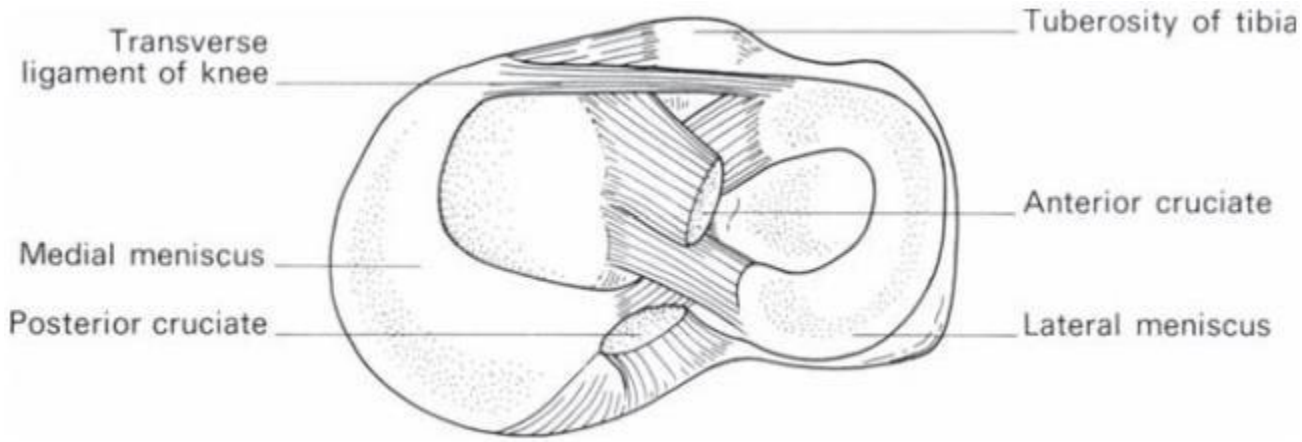
Clinical Anatomy

Applied anatomy for students and junior doctors

Harold Ellis

ELEVENTH EDITION

Diagram to show the upper surface of right tibia



Clinical Anatomy
Applied anatomy for students and junior doctors
Harold Ellis
ELEVENTH EDITION

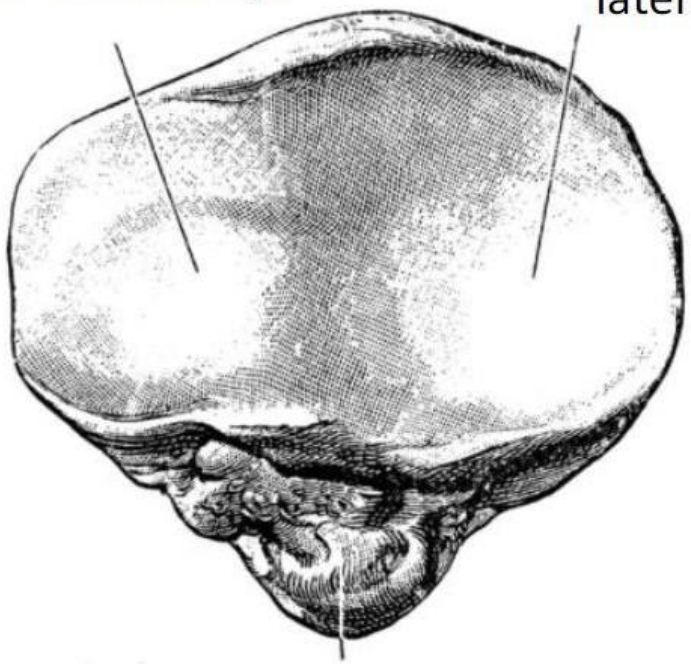
Right Patella



Quadriceps femoris

Vastus medialis

apex



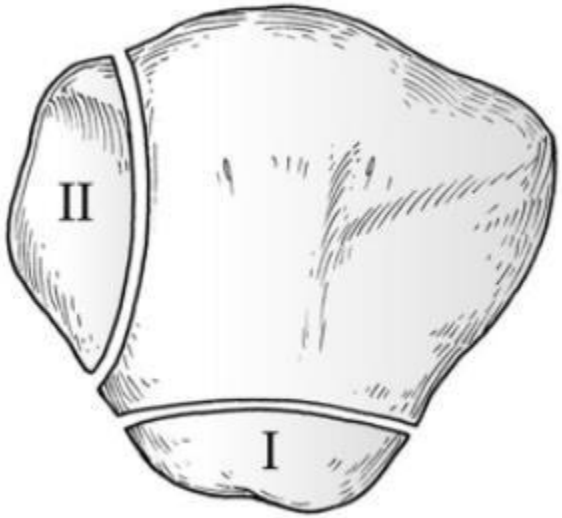
Facet for Medial condyle

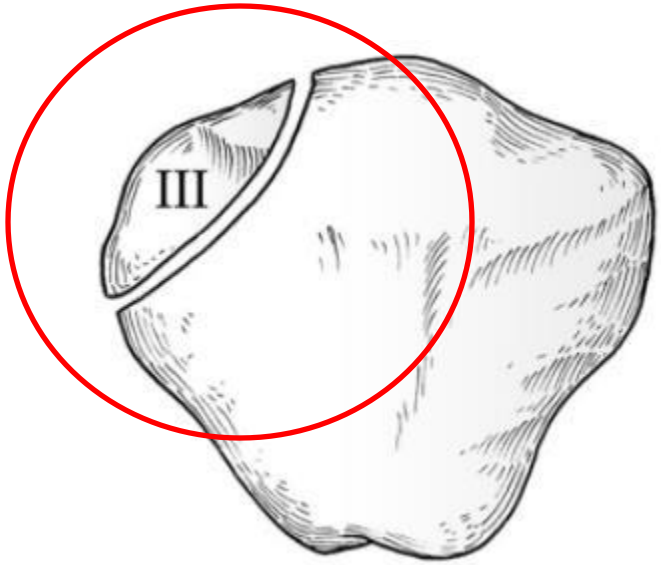
Facet for lateral condyle

Ligamentum patellae

Bipartite Patella
3 types

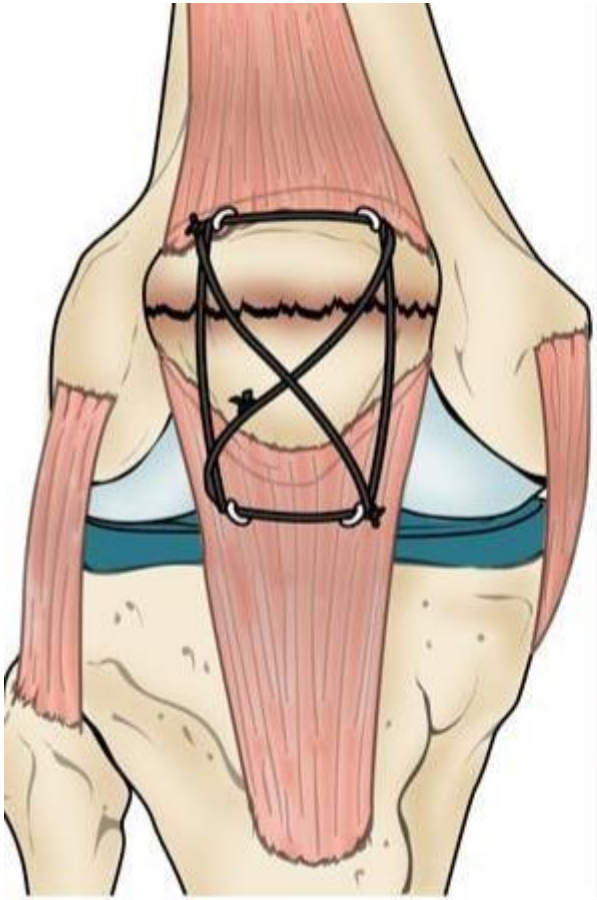
commonest





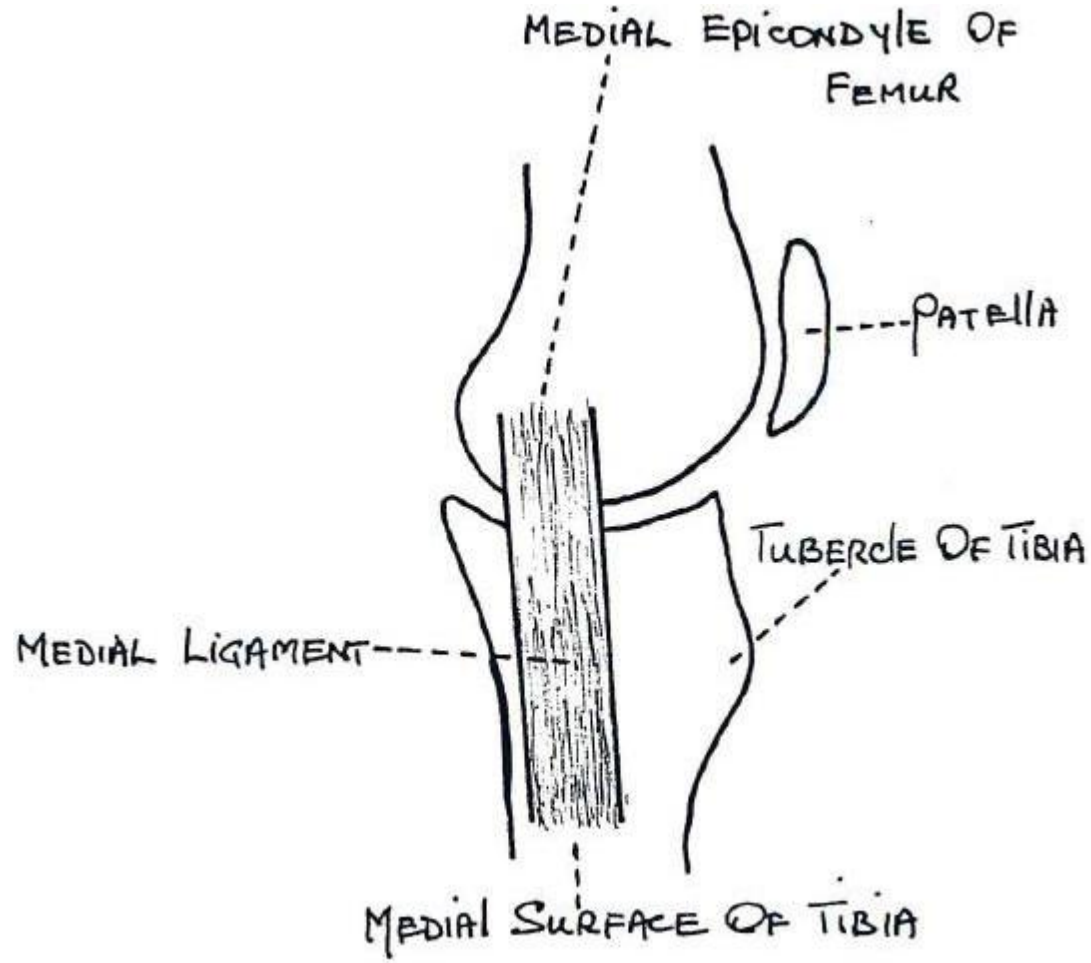
Saupe Classification		
Type	*Incidence*	*Location*
Type I	5%	Inferior pole
Type II	20%	Lateral margin
Type III	75%	Superolateral pole

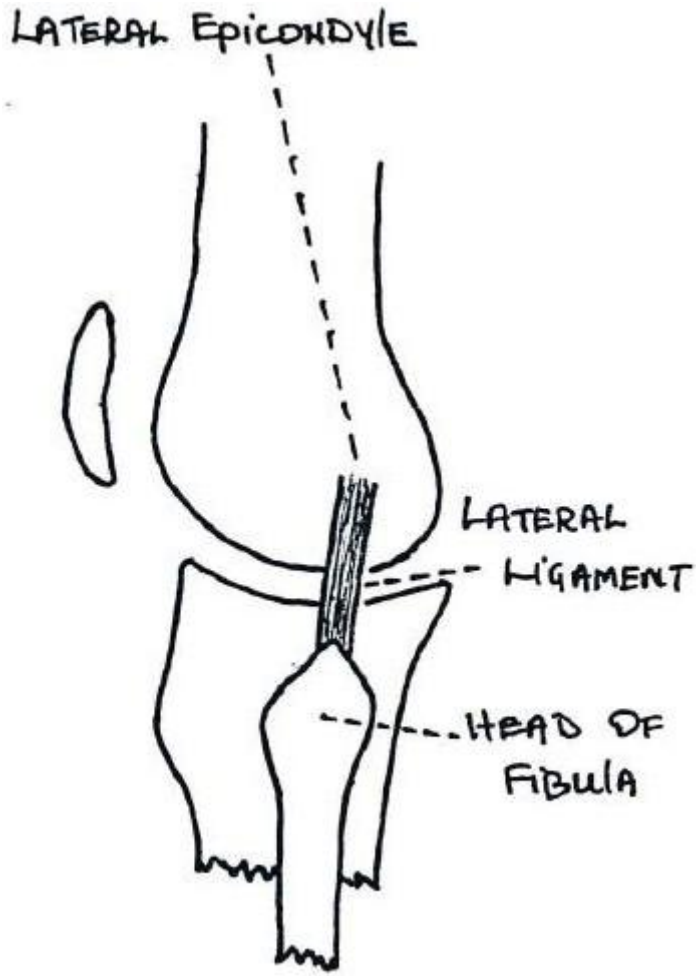
Treatment of transverse patella fracture



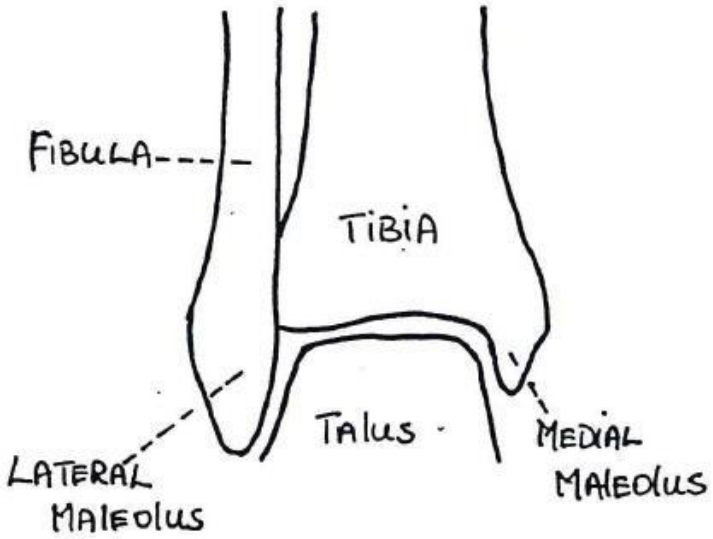


OrthoInfo

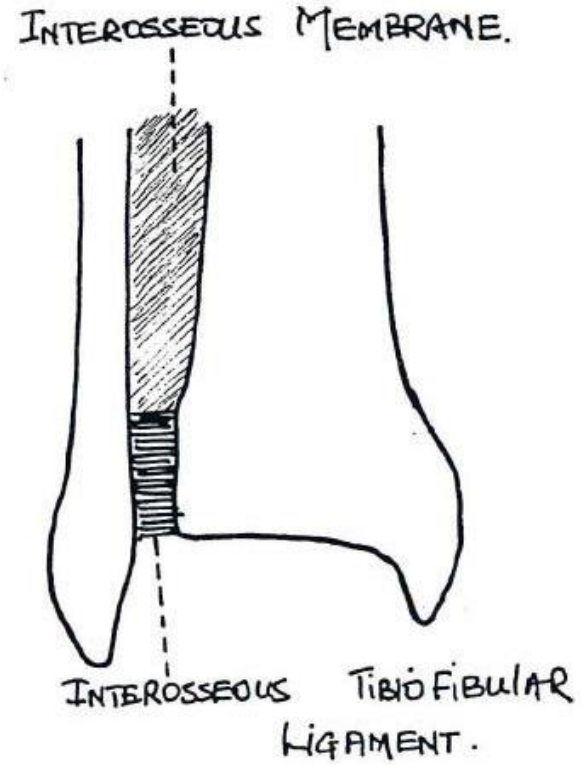

MEDIAL EPICONDYLE OF FEMUR
PATELLA
TUBERCLE OF TIBIA
MEDIAL LIGAMENT
MEDIAL SURFACE OF TIBIA


LATERAL EPICONDYLE
LATERAL LIGAMENT
HEAD OF FIBULA

Ankle joint is a simple hinge joint

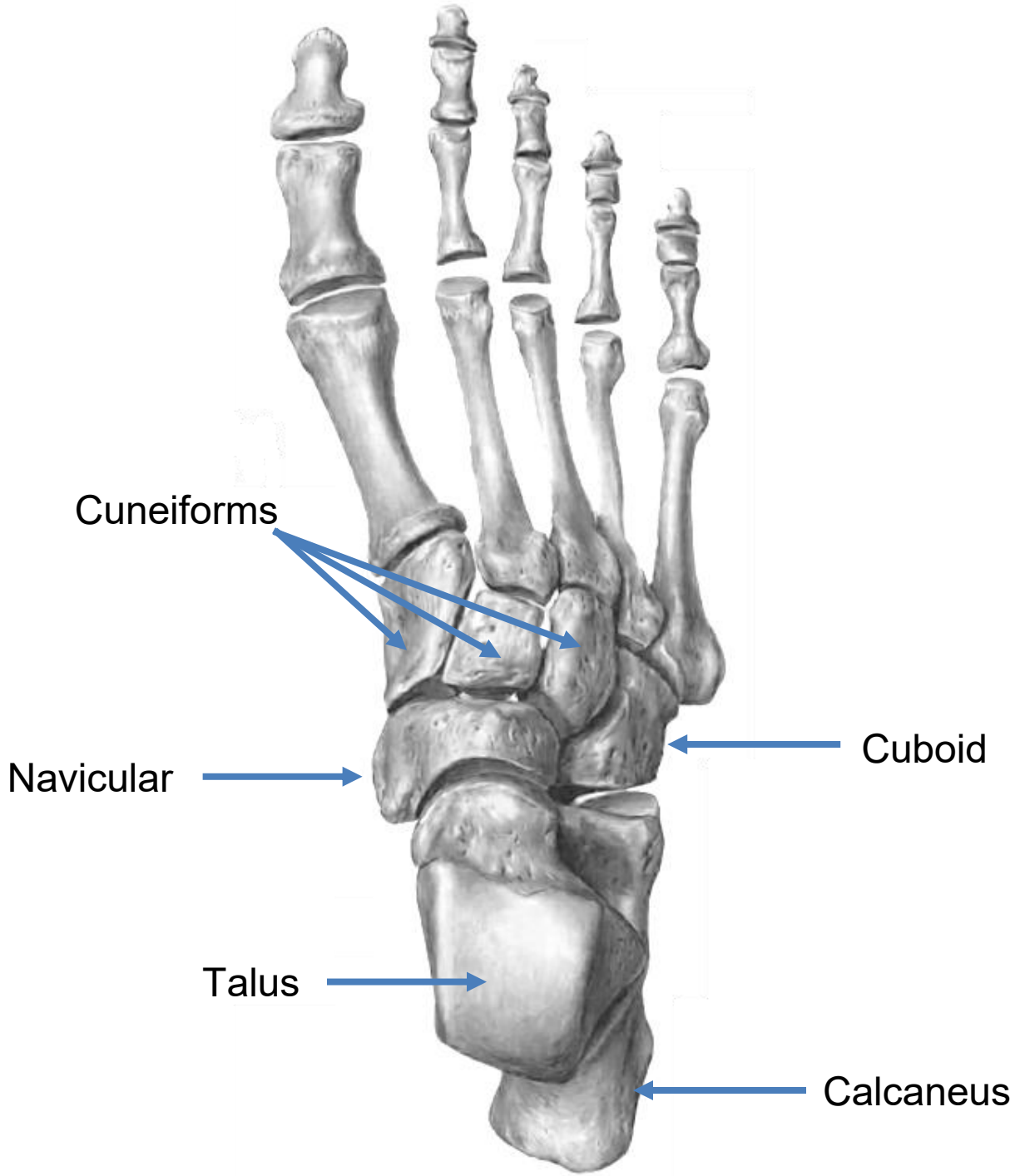


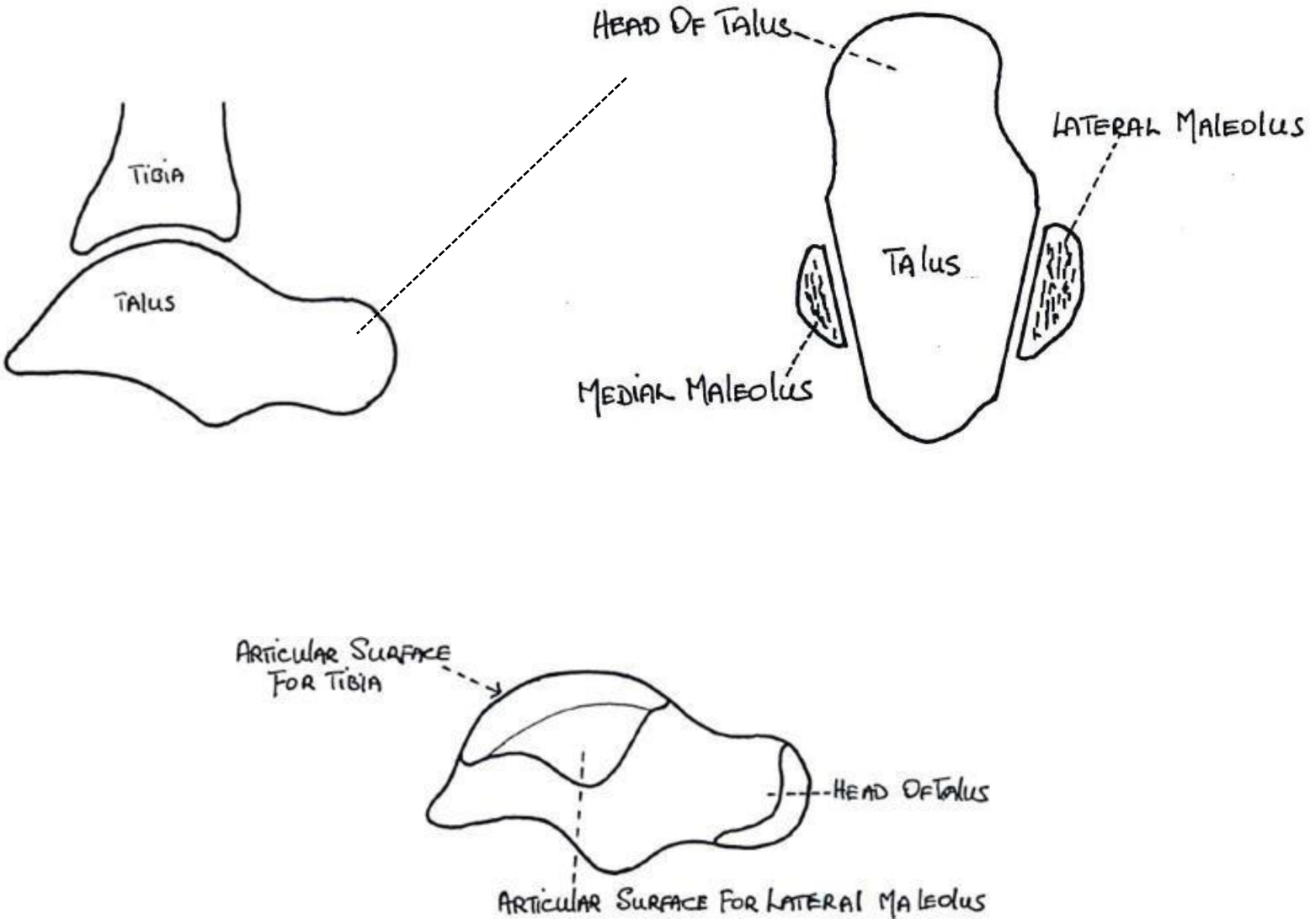
Inferior tibiofibular joint is a syndesmosis

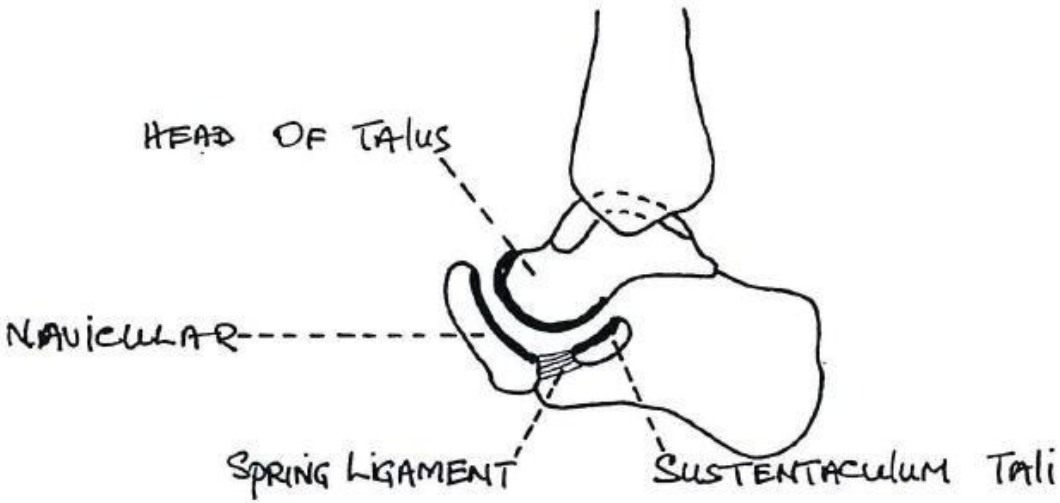


Syndesmosis is a fibrous joint with limited movement

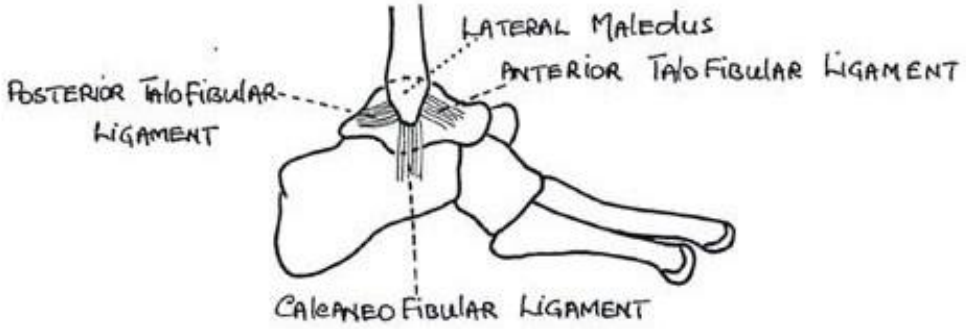
But it is a true joint with articular cartilage

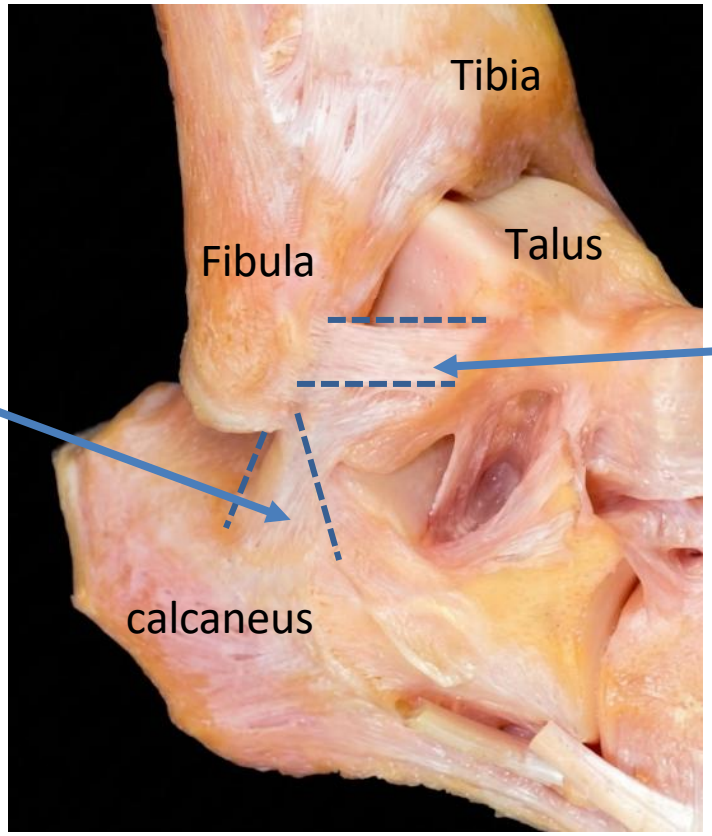

Cuneiforms
Navicular
Cuboid
Talus
Calcaneus


HEAD OF TALUS
LATERAL MALEOLUS
TIBIA
TALUS
TALUS
MEDIAL MALEOLUS
ARTICULAR SURFACE FOR TIBIA
HEAD OF TALUS
ARTICULAR SURFACE FOR LATERAL MALEOLUS


HEAD OF TALUS
NAVICULAR
SPRING LIGAMENT
SUSTENTACULUM TALI

Lateral view: ankle ligaments



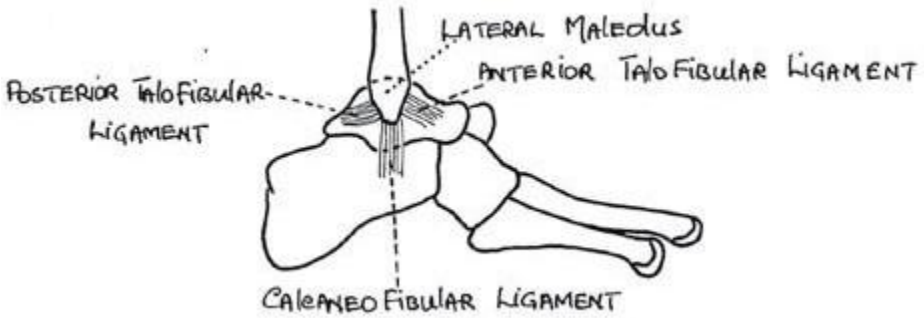


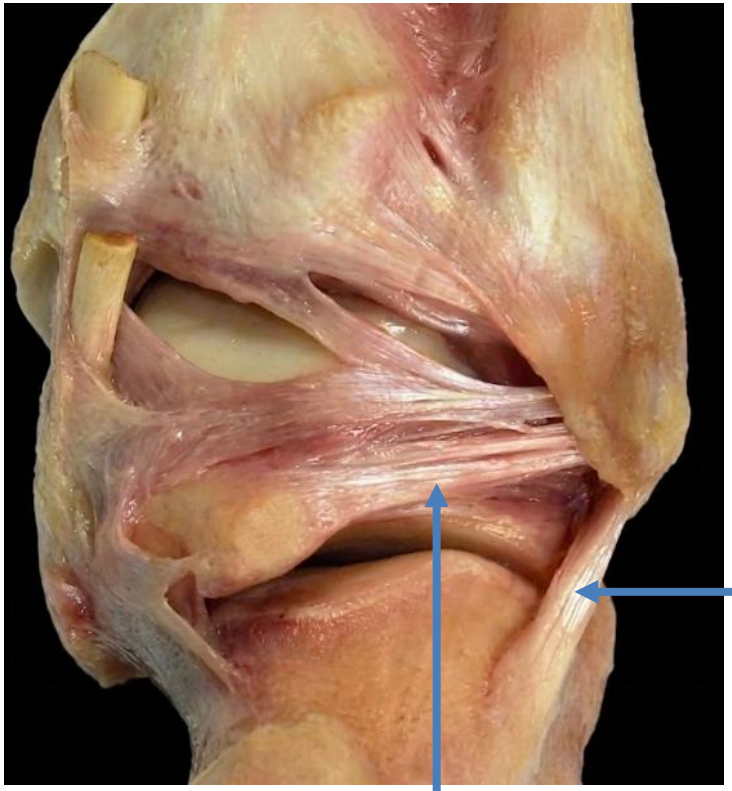
Calcaneofibular Ligament

Anterior talofibular Ligament

Golanó, P., Vega, J., de Leeuw, P.A.J. *et al.* Anatomy of the ankle ligaments: a pictorial essay. *Knee Surg Sports Traumatol Arthrosc* **18,** 557–569 (2010).

Posterior view: ankle ligaments





Calcaneofibular ligament

Posterior talofibular Ligament

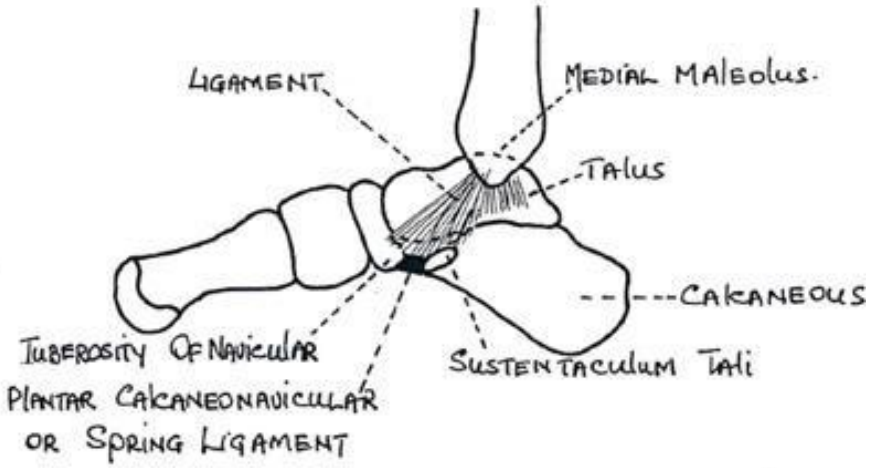
Knee Surg Sports Traumatol Arthrosc (2010) 18:557–569
DOI 10.1007/s00167-010-1100-x

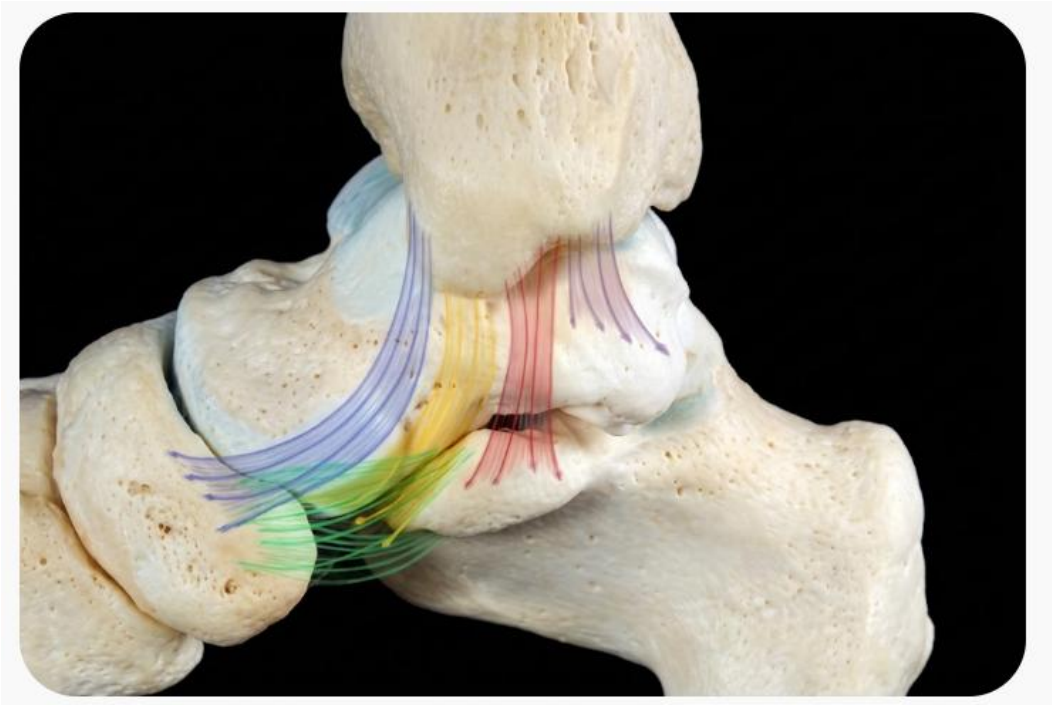
ANKLE

Anatomy of the ankle ligaments: a pictorial essay

Pau Golanó · Jordi Vega · Peter A. J. de Leeuw · Francesc Malagelada · M. Cristina Manzanares · Víctor Götzens · C. Niek van Dijk

Medial ligaments of ankle





Knee Surg Sports Traumatol Arthrosc (2010) 18:557–569
DOI 10.1007/s00167-010-1100-x

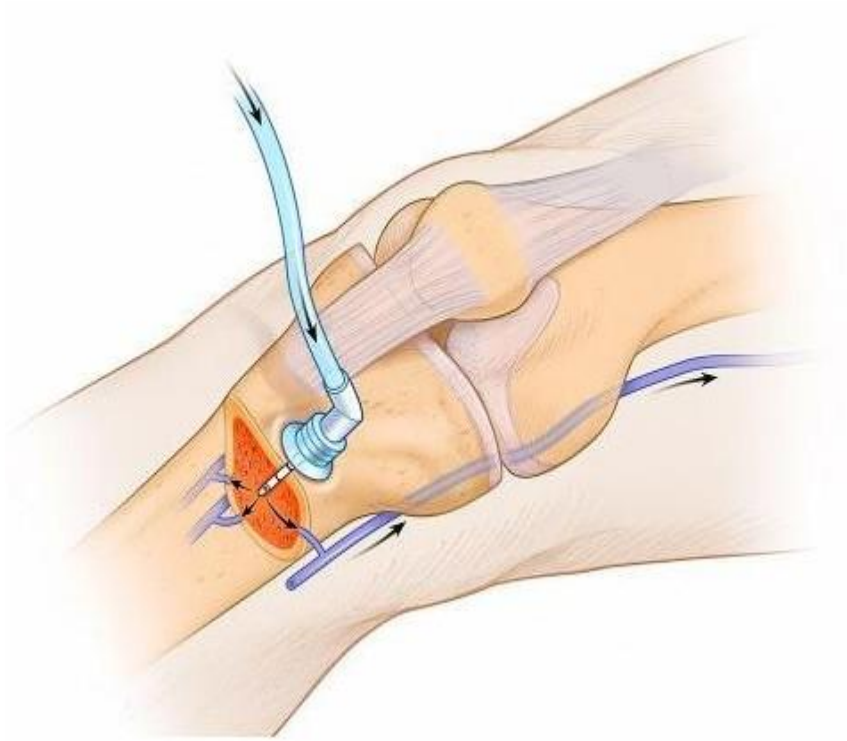
ANKLE

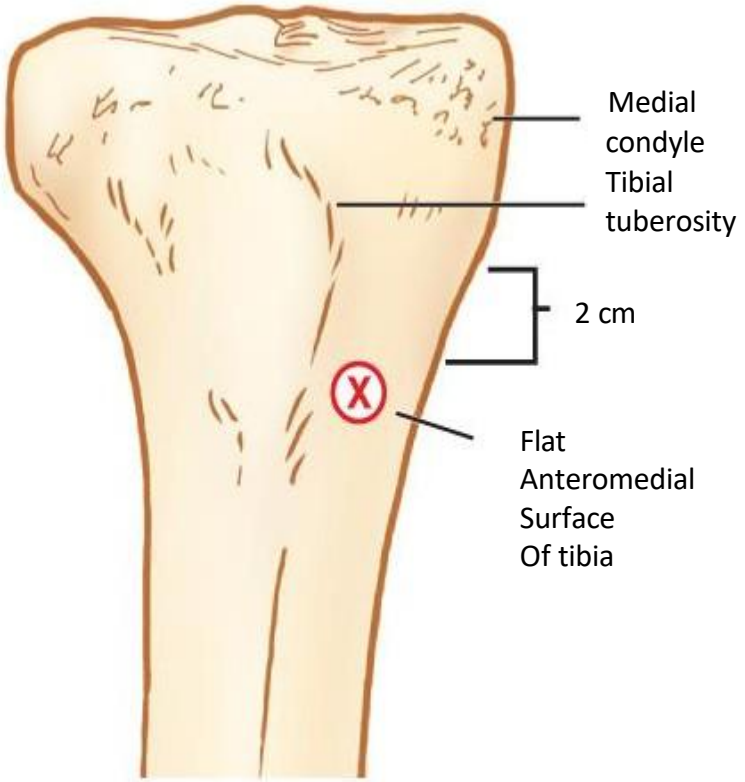
Anatomy of the ankle ligaments: a pictorial essay

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Clinical anatomy

Emergency Intraosseous infusion





Osgood-Schlatter disease

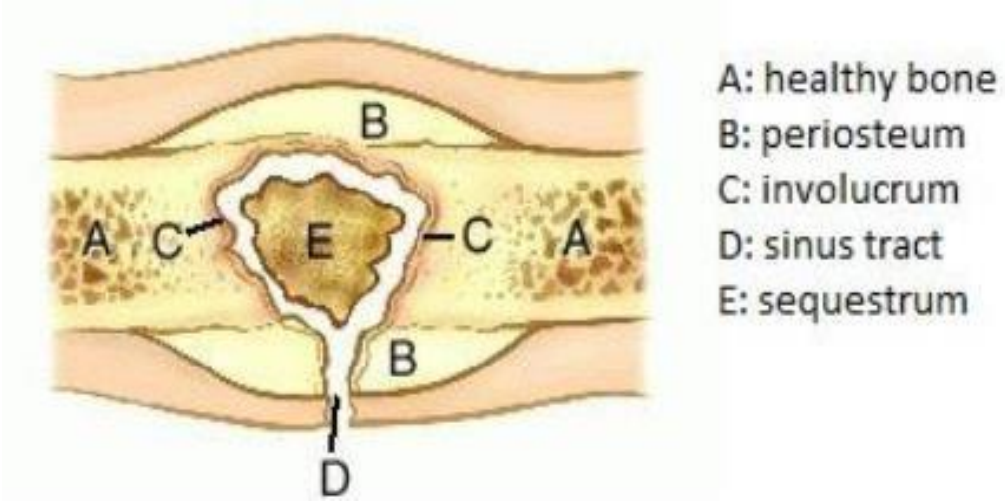


A disease of adolescent boys mainly

Acute Osteomyelitis is most commonly found in upper end of tibia



Causative organism is often: Staphylococcus aureus

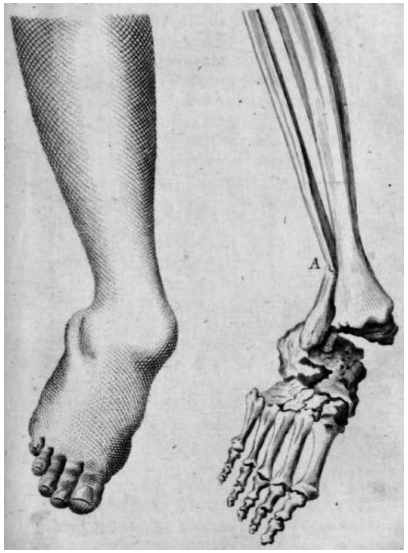


involucrum
: a layer of new bone

growth sequestrum
: necrotic bone which has become walled off from its blood supply and can present as a nidus for chronic osteomyelitis

Percival Pott (1713-1788) is famous for 4 contributions

- Potts' Fracture
- Pott's disease of spine (TB)
- Pott's puffy tumour
- Occupational cancer
  (scrotal cancer in child chimney sweeps)



Bi-malleolar fracture of ankle



TB of spine

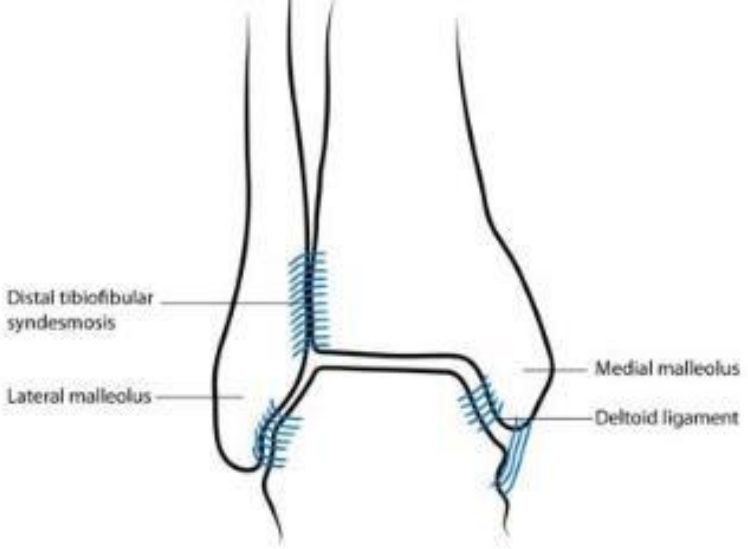


Osteomyelitis of frontal bone

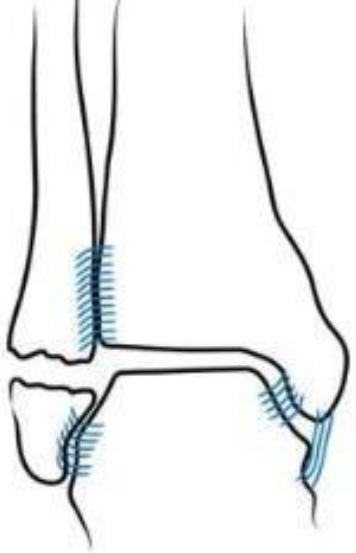


Scrotal carcinoma

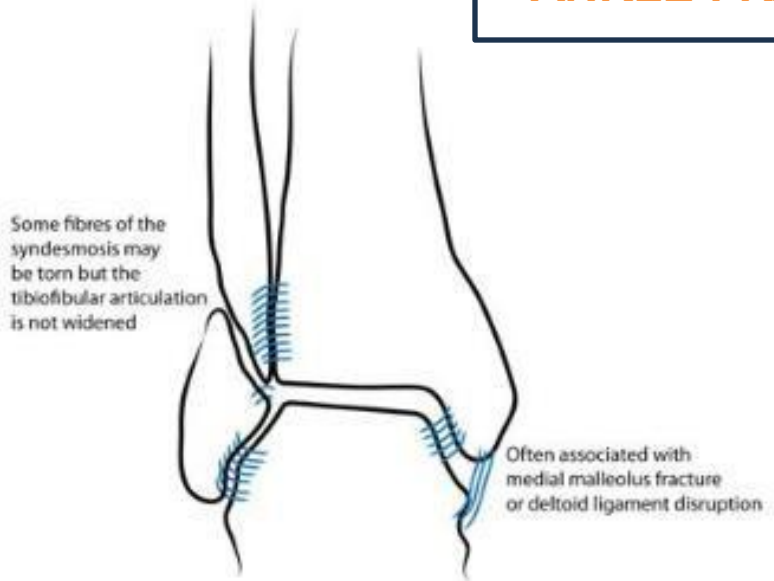
ANKLE FRACTURES



Normal



Weber A



Weber B



Weber C